

SMART SIGN LANGUAGE READER

25-26J-223

Project Proposal Report (Draft)

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Declaration

I declare that the work presented here is original and that we have not plagiarized in any way, shape, or form any portion of this proposal into another work that has already been submitted for a degree or diploma in any other university or higher educational institution without acknowledgement.

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The supervisor and co-supervisor should certify the proposal report with the following declaration.
The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor



Date

29/8/2025

Signature of the co-supervisor



Date

28/08/2025

Abstract

The Smart Sign Language Reader addresses the critical communication barrier faced by deaf and hard-of-hearing individuals in Sri Lanka, where existing SSL recognition systems rely primarily on hand gestures while overlooking essential non-manual features that convey contextual meaning and emotional nuance, with current technologies suffering from limited accuracy due to their uni-modal approach that fails to capture the multimodal nature of sign language communication integrating hand movements, facial expressions, and body posture simultaneously. This research implements a novel multimodal deep learning architecture that analyzes hand movements, facial expressions, and body posture simultaneously to achieve more accurate and context-aware interpretation of SSL compared to existing hand-gesture-only systems, employing advanced computer vision techniques including CNNs for spatial feature extraction and RNNs for temporal sequence modeling, combined with MediaPipe Holistic for real-time multimodal landmark detection and context-aware processing capabilities that differentiate between similar signs by understanding the situational context in which they are used. The Smart Sign Language Reader converts interpreted multimodal sign language inputs into meaningful Sinhala text output in real-time, enabling seamless communication between deaf and hearing individuals by incorporating facial expression analysis and body posture recognition alongside hand gesture detection to capture the full semantic richness of SSL communication, including grammatical markers and emotional context that are typically conveyed through non-manual features, addressing the significant gap in current SSL recognition systems that often produce incomplete or misleading interpretations due to their limited scope.

Keywords: Smart Sign Language, Multimodal, Emotions, Sinhala, MediaPipe

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LIST OF ABBREVIATIONS

Sinhala Sign Language	SSL
Convolutional Neural Networks	CNN
Artificial Intelligence	AI
Recurrent Neural Networks	RNN
Long Short-Term Memory	LSTM
Support Vector Machines	SVM
K-Nearest Neighbors	KNN

1. INTRODUCTION

Sign language recognition (SLR) has emerged as a critical research domain within computer vision and artificial intelligence, driven by the fundamental need to bridge communication barriers between deaf and hearing communities worldwide. The field has witnessed significant evolution from early rule-based systems to sophisticated deep learning approaches, with researchers focusing on both isolated word recognition and continuous sign language understanding [1]. For Sinhala Sign Language (SSL) specifically, research has been particularly challenging due to its classification as a low-resource language with limited labeled datasets and technological support. Early SSL research primarily focused on static fingerspelling recognition using traditional machine learning approaches such as Support Vector Machines (SVM) and Naive Bayes classifiers, achieving modest accuracy rates of 73.3% to 81.2% [2] [3] .

The development of an effective Smart Sign Language Reader requires mastery of several key technical domains and methodologies. Computer vision techniques form the foundational layer, particularly MediaPipe for real-time hand and pose estimation, which provides robust landmark detection capabilities essential for multimodal sign analysis [4] [2]. MediaPipe's pre-trained models can detect and track hands in real-time within video frames, identifying key points including fingertips, knuckles, and palm centers while estimating 3D positions of hand landmarks in 2D image frames. Deep learning architecture represents the core computational framework, with Convolutional Neural Networks (CNNs) serving as the primary mechanism for spatial feature extraction from visual input. Recent research emphasizes the importance of hybrid CNN-LSTM architectures for capturing both spatial and temporal dependencies in sign language sequences, with bidirectional LSTM networks proving particularly effective for analyzing sequential sign patterns [5] [6].

Current state-of-the-art sign language recognition systems have achieved remarkable performance improvements through advanced architectural innovations and large-scale dataset utilization [7] [1]. The most recent breakthrough includes Conformer-based architectures that adapt hybrid CNN-Transformer designs originally developed for speech recognition, achieving new benchmarks with parametric efficiency gains of 467-1807x reduction while maintaining competitive performance. Edge AI and lightweight deployment have become increasingly important, with recent models like Signformer achieving 2nd place on competitive leaderboards while operating with only 0.57 million

parameters, demonstrating that high performance can be maintained with significantly reduced computational requirements [8].

Previous approaches to sign language recognition have typically focused on uni-modal solutions that rely primarily on hand gesture analysis while neglecting crucial non-manual features [1] [4]. Traditional SSL recognition systems have employed static image classification methods using conventional machine learning algorithms like SVM, KNN, and Random Forest, achieving limited accuracy rates between 73% to 89%. Recent improvements have incorporated single-modality deep learning approaches using CNN architectures for spatial feature extraction and RNN/LSTM networks for temporal modeling, achieving better performance but still missing critical contextual information [5] [6].

Our Smart Sign Language Reader approach builds upon and varies from previous work in several key aspects. Unlike traditional uni-modal systems, our approach implements comprehensive multimodal integration that simultaneously analyzes hand movements, facial expressions, and body posture to capture the full semantic richness of SSL communication [4]. While existing multimodal approaches often process different modalities separately and perform late fusion, our system employs context-aware processing that differentiates between similar signs by understanding the situational context in which they are used, addressing a critical gap in current SSL recognition systems [1] [6]. Our approach specifically addresses the low-resource language challenge of SSL by implementing edge-optimized machine learning techniques that enable real-time processing on mobile devices without compromising accuracy [8]. Our system aims to achieve the optimal balance between comprehensive multimodal analysis and practical real-world applicability for the Sri Lankan deaf community.

1.1 Background and Literature Survey

Sign language recognition systems are technological solutions designed to automatically interpret and translate visual sign language gestures into text or speech output. These systems leverage computer vision, machine learning, and artificial intelligence to bridge communication barriers between deaf and hearing communities [9] [1]. In this literature review, we examine the historical development of sign language recognition in computer science and how multimodal approaches affect recognition accuracy and practical implementation.

Communication accessibility is a fundamental requirement for inclusive society, according to, but technological support for local sign languages like Sinhala Sign Language (SSL) remains severely limited in developing countries [4] [9]. Sri Lanka has a substantial deaf community that relies on SSL for daily communication, but the research indicates that there are numerous technical challenges to consider besides basic gesture recognition, such as facial expressions, body posture, and contextual understanding. The objective of previous studies has been to develop more sophisticated recognition systems for various sign languages. The approach suggested in this study is to implement a multimodal feedback system that simultaneously tracks hand movements, facial expressions, and body language to provide comprehensive sign language interpretation. For this methodology to be effective, the system must incorporate appropriate preprocessing and feature extraction techniques in advance.

The study focuses primarily on deep learning approaches for Sinhala Sign Language recognition over the past decade. Research demonstrates that CNN-based systems have achieved significant breakthroughs in static gesture recognition. Since traditional machine learning methods showed limited performance, deep learning architectures automatically extract features from visual input data. The goal of current research is to determine optimal network architectures and achieve higher recognition accuracy. Different approaches in sign language recognition depend on dataset quality, so data collection methodologies also have direct and indirect effects on system performance.

indicates that a major challenge in Sri Lanka, where approximately 466,000 people have hearing disabilities, is inadequate technological support for sign language communication. Researchers have struggled to develop comprehensive systems that capture the full complexity of SSL using either conventional computer vision methods or non-scientific approaches. Current systems often fail to consider crucial elements such as facial

expressions, body posture, temporal dynamics, and contextual meaning. This research proposes an accurate multimodal recognition system that incorporates all contributing factors, including hand gesture analysis, facial expression detection, body pose estimation, and contextual understanding. Additionally, modern artificial intelligence is enabling precision recognition systems, which are expanding in developing countries like Sri Lanka and focus on comprehensive multimodal analysis [10].

In, the emphasis is on technical factors specific to Sinhala Sign Language that can be utilized to address recognition challenges in assistive technology. Based on multimodal sensor information, sign language can be interpreted with greater accuracy. In sign language recognition, comprehensive feature extraction is critically important for achieving reliable performance.

The paper proposes developing an integrated multimodal sign language recognition system that analyzes gesture quality and provides accurate interpretation results stored in a comprehensive database. Various components in the system measure hand landmark positions, facial expression features, body pose coordinates, and temporal dynamics. After hand detection, facial analysis, pose estimation, and temporal modeling modules were established, a deep learning architecture for multimodal fusion had to be constructed. MediaPipe framework and CNN-LSTM hybrid models were utilized to create the system's feature extraction capabilities [2] [5]. The device enhanced communication accessibility significantly, helping deaf individuals achieve better social integration through more efficient technological solutions.

The integration of facial expressions is a crucial factor in sign language interpretation, and its systematic analysis should be incorporated to make better recognition decisions. Modern computer vision frameworks provide sensors for high-resolution real-time tracking of facial landmarks. These technologies are now commercially available in various applications. The objective of current studies is to evaluate how effectively multimodal approaches perform in real-world sign language scenarios. Readings from different modalities are compared with data from traditional single-modal recognition methods. Experiments are conducted in various configurations, including controlled laboratory tests, semi-controlled evaluations with static gestures, and field tests with dynamic continuous signing. Research indicates that additional investigation is required to determine optimal performance across different implementation contexts.

The literature shows that multimodal deep learning architectures, including CNN-LSTM hybrid models, are increasingly utilized in sign language recognition systems worldwide, both in high-resource and low-resource language environments. Multimodal approaches are computationally efficient, relatively straightforward to implement and optimize, and provide comprehensive gesture understanding by analyzing hand movements, facial expressions, and body posture simultaneously [5] [6]. Furthermore, multimodal systems offer numerous advantages, including improved accuracy for similar gestures, enhanced contextual understanding, and better handling of continuous signing sequences. Research demonstrates that implementing multimodal architectures can significantly improve recognition accuracy and system robustness while extending applicability to real-world scenarios. This advancement helps researchers achieve better performance outcomes. However, the success of multimodal systems depends on various factors, including computational resources, dataset quality, model architecture selection, and proper training methodologies [1] [7]. The authors emphasize that additional research is needed to determine optimal multimodal fusion strategies for different sign languages and implementation environments.

1.2 Research Gap

Current Sinhala Sign Language recognition systems predominantly rely on hand-gesture detection alone, neglecting the rich non-manual cues—facial expressions and body posture—that convey essential grammatical and contextual meaning in SSL. While some studies have explored pose estimation or CNN-LSTM hybrids for dynamic sign detection, they fail to integrate facial affect or full-body context, resulting in ambiguous or inaccurate translations when signs share similar hand configurations but differ in meaning. Moreover, existing solutions generally process video streams in the cloud, introducing latency, privacy concerns, and loss of responsiveness, and they rarely adapt to regional variations in SSL usage.

The proposed Smart Sign Language Reader addresses these deficiencies by:

- Simultaneously analyzing hand, face, and body movements to disambiguate signs with overlapping hand shapes.
- Incorporating contextual understanding through a multimodal deep-learning model that associates non-manual markers (e.g. eyebrow movements, head tilts) with syntactic and semantic features of SSL.

- Deploying the entire recognition pipeline on-device using lightweight, edge-optimized architectures (e.g. TensorFlow Lite with MediaPipe), ensuring low latency, user privacy, and offline functionality.
- Adapting to regional and dialectal variations of Sinhala Sign Language via transfer learning on a culturally diverse, annotated dataset.

1.3 Research Problem

Current Sinhala Sign Language recognition systems predominantly rely on hand-gesture analysis, lacking integration of facial expressions and body posture to capture full communicative intent. While many solutions achieve moderate accuracy with single-modality models, few provide real-time, on-device processing or context-aware interpretation that distinguishes similar signs based on situational cues. This gap limits the expressiveness and reliability of SSL translation tools, hindering seamless communication between deaf and hearing users and reducing the practical applicability of digital sign language assistants in everyday, dynamic environments.

Furthermore, most existing systems focus on word-level or character-level recognition, failing to address the complexity of sentence-level, continuous signing that reflects natural communication practices among Sri Lanka's deaf community. Limitations in data quality, variability among signers, and difficulty in collecting large, annotated multimodal datasets have slowed progress, resulting in inconsistent accuracy and poor generalization in diverse real-world scenarios. Usability challenges such as environmental noise, inadequate robustness in varying lighting conditions, and dependence on external hardware (like gloves or sensors) further restrict adoption. Comprehensive solutions remain elusive due to insufficient integration of contextual and non-manual features in existing models, leaving a significant gap in accessible, inclusive digital communication technologies for the Sri Lankan deaf community.

2. OBJECTIVES

2.1 Main Objective

To develop a real-time, on-device Smart Sign Language Reader that integrates multimodal deep learning for hand gesture recognition, facial expression analysis, and body posture estimation, enabling accurate and context-aware translation of Sinhala Sign Language into text, thereby enhancing communication effectiveness and inclusivity for deaf users in everyday environments.

2.2 Sub Objectives

1. Implement a robust computer vision pipeline using MediaPipe Holistic to detect and track hand landmarks, facial landmarks, and body pose keypoints in live video streams with low latency.
2. Design and train a convolutional neural network (CNN) for spatial feature extraction from hand, face, and pose inputs, and integrate it with a bidirectional LSTM network for temporal sequence modeling of continuous sign language gestures.
3. Develop a contextual interpretation module that fuses multimodal feature embeddings to differentiate between similar signs based on grammatical markers, signer intent, and situational context.
4. Optimize the entire model for on-device deployment using TensorFlow Lite and quantization techniques to ensure real-time performance and privacy-preserving operation on mobile platforms.
5. Evaluate system performance on real-world test sets, measuring recognition accuracy, latency, and user satisfaction in educational and public service scenarios.

3. METHODOLOGY

This section describes the workflow, tasks, materials, data requirements, and timeline for developing the Smart Sign Language Reader component.

3.1 System Architecture and Workflow

3.1.1 System Diagram

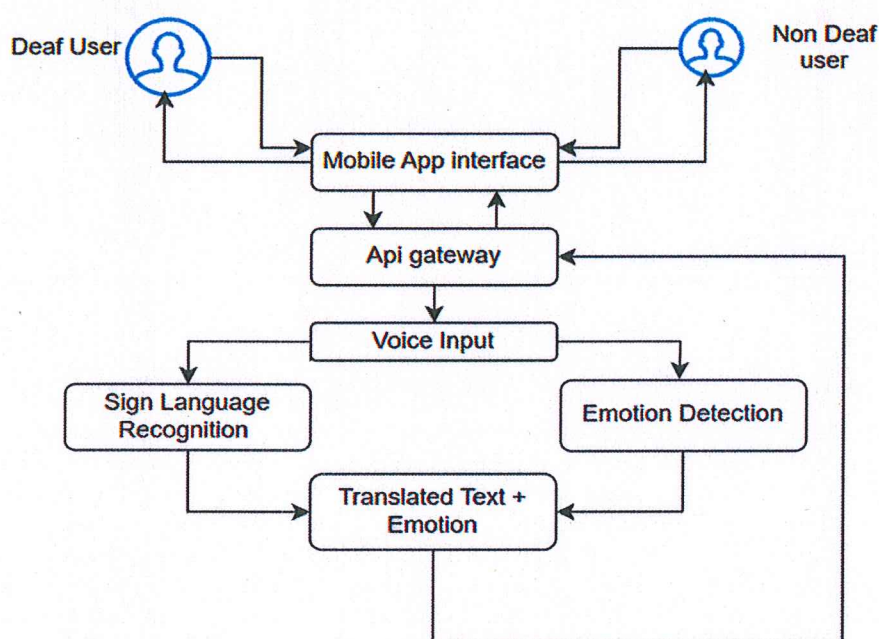


Figure 3-1 System Diagram

3.1.2 System Workflow

1. Capture: Acquire live video input from the device camera.
2. Preprocessing: Apply MediaPipe Holistic to detect and normalize hand, face, and body landmarks.
3. Feature Extraction: Use a CNN to extract spatial features from each modality.
4. Temporal Modeling: Feed feature sequences into a bidirectional LSTM to learn sign language dynamics.

5. Contextual Fusion: Fuse multimodal embeddings and apply a context-aware disambiguation layer.
6. Output: Translate fused features into Sinhala text and render text-to-speech output.
7. Logging: Record recognition events, confidence scores, and timestamps in the on-device database.

3.2 Data Requirements and Collection

- SSL400 dynamic dataset for baseline model training.
- Augmented recordings collected from 20 native SSL users to capture signer variability.
- Annotation schema for sign glosses, facial expressions, and body posture tags.
- Collection method: Controlled studio recordings and field captures in realistic environments.

3.3 Tasks and Subtasks

1. Data Collection & Annotation

- Curate and label an SSL multimodal dataset (video recordings of hand gestures, facial expressions, and body movements).
- Annotate sign boundaries, glosses, and contextual metadata.

2. Model Development

- Implement MediaPipe Holistic pipeline for real-time landmark extraction.
- Design and train CNN architecture for spatial feature extraction.
- Develop and train bidirectional LSTM for temporal modeling.
- Create context-aware fusion module to integrate multimodal embeddings.

3. On-Device Optimization

- Quantize and convert the trained model using TensorFlow Lite.
- Integrate the model into a cross-platform mobile application.

4. Application Development

- Develop UI components for live recognition, playback, and log review.
- Implement text-to-speech engine for rendered translations.
- Add settings for language, model selection, and logging controls.

5. Testing & Evaluation

- Conduct unit tests for each module.
- Perform end-to-end testing on Android and iOS devices.
- Evaluate performance metrics: accuracy, latency, model size, and user satisfaction.

3.4 Materials and Resources

- Equipment: Smartphones with front-facing cameras, development workstation (GPU-enabled).
- Software: Python, TensorFlow, TensorFlow Lite, MediaPipe, React Native, SQLite, Git (Version Controlling)
- Graphics: UI wireframes and icon assets.
- Others: SSL400 dataset, additional recorded sign language videos, annotation tools.

3.5 Anticipated Outcomes

- A real-time mobile application that accurately translates SSL gestures into Sinhala text and speech.
- Model performance metrics: $\geq 90\%$ recognition accuracy, ≤ 200 ms latency
- Deployment-ready software with user-friendly interface and offline operation capability.
- Potential real-world applications in education, public services, and personal communication, empowering greater inclusivity and accessibility for the Sri Lankan deaf community.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system shall capture live video input from the device camera and perform real-time detection of hand landmarks, facial landmarks, and body pose keypoints.
- The system should preprocess and normalize multimodal inputs before feeding them into the trained CNN-LSTM recognition pipeline.
- The system shall translate recognized SSL gestures into corresponding Sinhala text with $\geq 90\%$ accuracy.
- The system shall provide context-aware disambiguation for similar signs by analyzing temporal sequences and multimodal cues.
- The system shall support export of translated text to speech via an integrated Sinhala text-to-speech engine.
- The system shall log recognition events, timestamps, and confidence scores to a local on-device database for user review and model improvement.
- The system should operate fully on-device without requiring internet connectivity to ensure privacy and low latency.

4.2 Non-Functional Requirements

- The user interface shall be intuitive and accessible, adhering to WCAG 2.1 AA standards for visual clarity and ease of use.
- The application shall be cross-platform compatible, supporting Android and iOS devices.
- Recognition latency from camera capture to translated text display shall not exceed 200 ms per frame.
- The on-device model size shall not exceed 50 MB to ensure feasibility on mid-range mobile hardware.
- The system shall achieve at least 90% uptime in continuous operation and include error-handling mechanisms for camera and sensor failures.

5. GANTT CHART

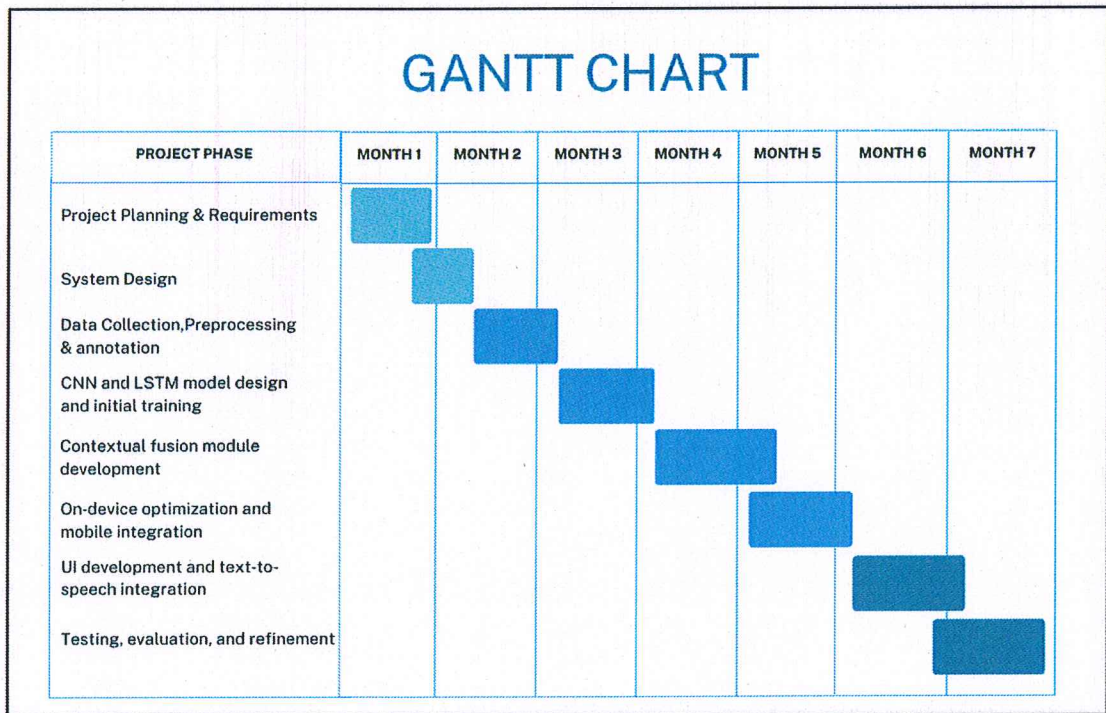


Figure 5-1 Gantt Chart

6. BUDGET AND BUDGET JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Cloud GPU Services (Training)	35,000	For training deep learning models.
Mobile Device for Testing	40,000	For validating real-time performance and testing.
Miscellaneous	5,000	To cover unexpected expenses, such as software tools, or unforeseen development costs that may arise during the project.

Table 6-1 Budget Estimation Table

Total Estimated Budget: LKR 80, 000

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